The other tiny hope for natural wormholes is the big bang creation of the universe. It is conceivable, but very unlikely, that traversable wormholes could have formed in the big bang itself. Conceivable for the simple reason that we don't understand the big bang well at all. Unlikely because nothing we do know about the big bang gives any hint that traversable wormholes might form there.

Can Wormholes Be Created by an Ultra-Advanced Civilization?



An ultra-advanced civilization is my only serious hope for making traversable wormholes. But it would face huge obstacles, so I'm pessimistic.

One way to make a wormhole, where previously there were none, is to extract it from the quantum foam (if the foam exists), enlarge it to human size or larger, and thread it with exotic matter to hold it open. This seems like a pretty tall order, even for an ultra-advanced civilization, but perhaps only because we don't understand the quantum gravity laws that control the foam, the extraction, and the earliest stages of enlargement (Chapter 26). Of course, we don't understand exotic matter very well either.

At first sight, making a wormhole seems easy (Figure 14.8). Just push a piece of our brane (our universe) downward in the bulk to create a thimble, fold our brane around in the bulk, tear a hole in our brane just below the thimble, tear a hole in the thimble itself, and sew the tears together. *Just!*